DCR3b: LLM Counterfactual Dependence Also Puts T1 Third — And Now We Know Why

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Date-Cut Retrodiction — DCR3b (intervention-algebra reframe test) **Status:** Overall **NO_GO**. M1 NO_GO (T1 rank 3 at 1904). M2 GO. M3 auto-NO_GO. M4 GO. The intervention-algebra reframe applied as LLM-based counterfactual dependence *within the corpus* also fails to reach T1. The failure is informative and consistent with DCR3: **whether you measure frequency or LLM-judged counterfactual weight, T2 (aether frame) always outscores T1 (simultaneity), because T2 is explicitly counterfactual-dependent in the corpus's own reasoning and T1 is not. Date:** 2026-07-27

Abstract

After DCR3 rejected the DR-arc's corpus-frequency nominator, the human director proposed a reframe: the object of nomination lives in the **intervention algebra**, not the corpus statistics. DCR3b tests the DCR-arc version of that reframe: replace the DR3 scoring's degree(p) (document count for content-equivalent propositions) with counterfactual_dependence(p) — an LLM-judged score for "how many other propositions in the same cut would become false or underdetermined if this one were false." Everything else (target_v4 classification, multidoc aggregation, ground truth T1 class, random null) identical to DCR3.

Three sandboxed Claude subagents per cut. 254 propositions at the 1904 target cut, 135 at 1897, 51 at 1880. Consensus by median.

Result: NO_GO. T1 still ranks third at 1904 under counterfactual scoring:

rank	class	LLM-counterfactual score	DCR3 (frequency)
1	unclassified	725	302
2	T2_privileged_frame	39	15
3	**T1_absolute_simultaneity**	**20**	**9**
4	T3_local_time_artifice	6	2

The rank order is preserved from DCR3. Both metrics agree: T2 (aether frame) is more counterfactually load-bearing in the corpus's own reasoning than T1 (simultaneity). The LLM judgment isn't wrong — it's *correct* about the corpus. Removing "the ether exists" cascades through nearly every proposition. Removing "there's a common time t" doesn't cascade through any proposition explicitly, because no proposition in the corpus cites it as an explicit premise.

That's the deep finding of DCR3b: the intervention-algebra reframe applied *within-the-corpus* fails for the same structural reason DCR3 failed. Silent commitments are silent about their own counterfactual weight too. The LLM asked "what depends on this?" looks at the same corpus and finds the same answer: T2 has explicit downstream consequences, T1 doesn't. Einstein's deletion was available precisely because T1 had shed its explicit counterfactual traces — no derivation *stated* that its validity required absolute simultaneity, so no in-corpus counterfactual measure can weight it correctly.

Third serial preregistered null on the DR-arc side (DCR3, DCR3b, and implicit in DCR2a). The pattern: **every scoring method that operates on the corpus's own visible content fails to reach the deletion.** The reframe's abstract idea is right — the object is in counterfactual dependence — but the operationalisation must be on the implicit argument structure, not the explicit propositional content.

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1. What was preregistered

DCR3B_PREREGISTRATION.md (2026-07-27, before run_dcr3b.py was drafted, before subagent calls were spawned) fixed:

runner: c5440337f71ebe54f5a941284831ea3c9bc47351d0a4045e0cc46229d97291b4.

each other and to labels.

multidoc(min_docs=2) aggregation, T1 ground truth, 10,000 random-null permutations at seed 20260727.

(M1 beats random null at p < 0.01), M4 (prompt committed).

- **Scoring**: score(p) = counterfactual_weight(p), LLM-judged.
- **Prompt** committed to DCR3B_PROMPT.md, SHA-256 pinned in the
- **Verifiers**: three sandboxed Claude subagents per cut, blind to
- **Consensus**: median across three.
- **Everything else identical to DCR3**: target_v4 classification,
- **Gates**: M1 (T1 first at 1904), M2 (T1 not first at 1880), M3

Nothing tuned after results.

2. Results

1904 (target cut, 254 propositions):

rank	class	LLM-counterfactual score	members	documents
1	unclassified	725	235	15
2	T2_privileged_frame	39	11	8
3	**T1_absolute_simultaneity**	**20**	7	5
4	T3_local_time_artifice	6	2	2

1880 (deep placebo, 51 propositions): T1 has 1 member from 1 doc → multidoc score 0. T1 rank position 2 in the raw output (alphabetic tiebreak among 0-score classes), consistent with DCR3.

Verifier stability: the three LLM verifiers agreed strongly on which propositions were high-dependence (ether-fills-space, propagation finite velocity, ether-at-rest-relative-to-earth all scored 8-10 by all three verifiers). No verifier disagreed by more than 3 points on any load-bearing proposition. Consensus by median is stable.

3. Gate decisions

gate		
M1 T1 first at 1904	**NO_GO**	rank 3
M2 T1 not first at 1880	GO	multidoc de-ranks singleton
M3 beats random null	NO_GO	by construction (M1 failed)
M4 prompt committed	GO	SHA-256 pinned

Overall NO_GO.

Licensed reading (from the runner):

> *counterfactual_scoring_still_missed_T1: even LLM-based*
>
> *counterfactual dependence puts T1 at rank 3. Third serial null on*

the intervention-algebra reframe on this arc.

4. Why the LLM scoring didn't rescue T1

The LLMs agreed with the frequency scorer that T2 outweighs T1. That agreement is the finding.

Consider the top counterfactual-weight propositions at 1904, per verifier consensus:

- Maxwell's ethereal_medium_fills_space (9–10)
- Michelson's ether_at_rest_earth_moves (7–9)
- Michelson's ether_fills_all_space (8–9)
- Larmor's absolute_space_is_aether (7–8)
- Maxwell's propagation_finite_velocity (7–8)

All of these are T2 (aether-frame) commitments. Every one is explicitly cited by other propositions in the corpus — Michelson- Morley's derivation depends on Maxwell's ether-fills-space; Lorentz's 1904 paper depends on Larmor's absolute-space-as-aether; every null-result argument depends on the ether being at rest relative to some frame. The LLMs correctly identified these as counterfactual-load-bearing because they *are*.

Now consider the T1 propositions (5 documents, 7 members):

- Larmor's common_time_across_two_systems
- Lodge's time_of_journey_perfectly_definite
- Maxwell's instant_across_whole_medium
- Poincaré's eclipse_perceived_simultaneously_over_earth
- Poincaré's regarded_as_simultaneous
- Lorentz's corresponding_instants
- Poincaré's same_causes_same_time

Each is a genuine simultaneity commitment. But *no other proposition in the corpus cites any of them as a premise*. Michelson-Morley doesn't say "and by the way this requires absolute simultaneity." It says "the time required for light to pass from one point to another depends on the direction," treating "the time required" as if it were a well-defined operational quantity. Larmor's derivation of the transformation doesn't say "presupposing a common time t"; it introduces t as a variable and uses it. Poincaré 1898 discusses simultaneity as a *philosophical* topic without ever citing it as a premise for another physical claim in the corpus.

The counterfactual scorer measures dependence in the corpus's *explicit* argument structure. T1 propositions live outside that structure — they are presupposed globally rather than cited locally. So the counterfactual scorer correctly reports low dependence.

The intervention-algebra reframe's abstract claim is correct — "the object is in counterfactual dependence, not in frequency" — but the operationalisation as LLM-scored within-corpus dependence is constrained by the same structural feature that made T1 invisible in the first place. To reach T1, you'd need counterfactual scoring on the *implicit* argument structure — inferring "what would this derivation require to be true even if the derivation doesn't say so?" That's a much stronger inference task than DCR3b's LLM performed, and it collides directly with DR7's soundness-completeness gap on open-realisation g functions.

5. What DCR3b does not license

operationalisation of one specific claim on one specific corpus. Other operationalisations may work.

on the propositions it can see, it correctly identifies counterfactual dependence. The failure is that the target commitment is systematically absent from the propositions it can see.

intervention-algebra reframe, in its within-corpus form, doesn't reach it. A stronger operationalisation (implicit-argument- inference) remains untested.

- **The reframe is refuted in general.** DCR3b tested one specific
- **LLM counterfactual scoring never works.** It works fine here —
- **The DR-arc is unable to identify T1.** DCR3b only shows the

6. Next work

in the corpus (identified by a subagent), ask the subagent to list the premises the derivation *requires to be valid*, whether or not those premises are stated. Score each proposition by how often it appears in inferred premise sets. This is DR7's semantic-access-to- D condition: the LLM has to reason about what the derivation requires, not what it says. Predicted-in-principle by DR6/DR6e to work when Claude has real semantic access; predicted-in-principle by DR7 to have its own g-soundness issues.

- **DCR3c** — implicit-argument-inference scoring. For each derivation

argument structure distinction: prove that any scoring function restricted to explicit premises has provably worse identification than one with access to inferred premises. Would generalise DCR3 + DCR3b into a specific corollary.

- **Theorem DR9** — formalise the corpus-explicit vs implicit

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Appendix: reproduction

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uv run --no-sync python -m experiments.date_cut_retrodiction.run_dcr3b
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Reads nine scores_<YEAR>_<A/B/C>.json files from results/dcr3b/, aggregates by median, applies same class assignment (target_v4) and multidoc gating as DCR3, computes M-gates, writes results/dcr3b_verdict.json. Local CPU, seconds.

Preregistration digest (SHA-256 of DCR3B_PREREGISTRATION.md**):** 34ab193dc2766a1ac90314c47b7ce2868ebb927cfc1f7401154d669aee6801e1.

Prompt digest (SHA-256 of DCR3B_PROMPT.md**):** c5440337f71ebe54f5a941284831ea3c9bc47351d0a4045e0cc46229d97291b4.